

Project 3: Network Intrusion Detection, Protocol Forensics & Dual-SIEM Threat Hunting

Engineering Challenges & Root Cause Analysis

1. Executive Summary & Architecture

As adversaries are increasingly leveraging fileless malware and living-off-the-land (LOTL) techniques to evade endpoint security, robust network traffic analysis (NTA) will become a critical fail-safe. The objective of this project was to architect a dedicated Network Intrusion Detection System (NIDS), engineer custom signatures for volumetric network anomalies, and correlate live packet captures with endpoint telemetry across a Dual-SIEM environment.

This project demonstrates the ability to intercept raw Layer 2 traffic via promiscuous sniffing, dissect TCP session states at the packet level, and translate unindexed wire data into actionable threat hunting intelligence using Splunk (SPL) and Wazuh.

2. Target Role & Skills Matrix

Target Role: Cybersecurity Analyst / Detection Engineer

Demonstrated Technical Skills:

- **Network Forensics & Protocol Analysis:** Packet dissection, TCP flag manipulation (`SYN` vs `ACK`), connection state tracking, Wireshark `.pcap` analysis.
- **Detection Engineering:** AF_PACKET promiscuous sniffing, Emerging Threats (ET) rulesets, authoring custom threshold-based Suricata signatures.
- **Dual-SIEM Operations:** Wazuh EVE JSON ingestion pipelines, Splunk Enterprise containerization, structured XML parsing (`KV_MODE`).
- **Threat Hunting (SPL Analytics):** Volumetric anomaly detection (Top Talkers), Long-tail outlier analysis (Rare Processes), Authentication tracking (Windows Event 4624/4625).
- **Linux Systems Administration:** Docker container deployment, Logical Volume Management (LVM) disk expansion, daemon troubleshooting.

3. Architectural Topology

Plaintext

```
[Kali Linux (Attacker)] --- (192.168.56.103)
      |
      v
[VirtualBox Host-Only Switch]
      |---> (Promiscuous SPAN/Tap) ---> [Ubuntu 24.04 (Sensor & SIEM Node)]
      |                                  ├── Suricata NIDS (AF_PACKET on enp0s8)
      |                                  ├── Wazuh Agent (Tailing eve.json)
      |                                  └── Splunk Enterprise (Docker)
      v
[Windows Server 2019 (DC01)] --- (192.168.56.110)
      ├── Services: DNS (53), Kerberos (88), SMB (445), LDAP (389)
      └── Telemetry: Sysmon (Event 1) & Security (Event 4624/4625)
```

Phase 1: Adversarial Emulation (MITRE ATT&CK T1046)

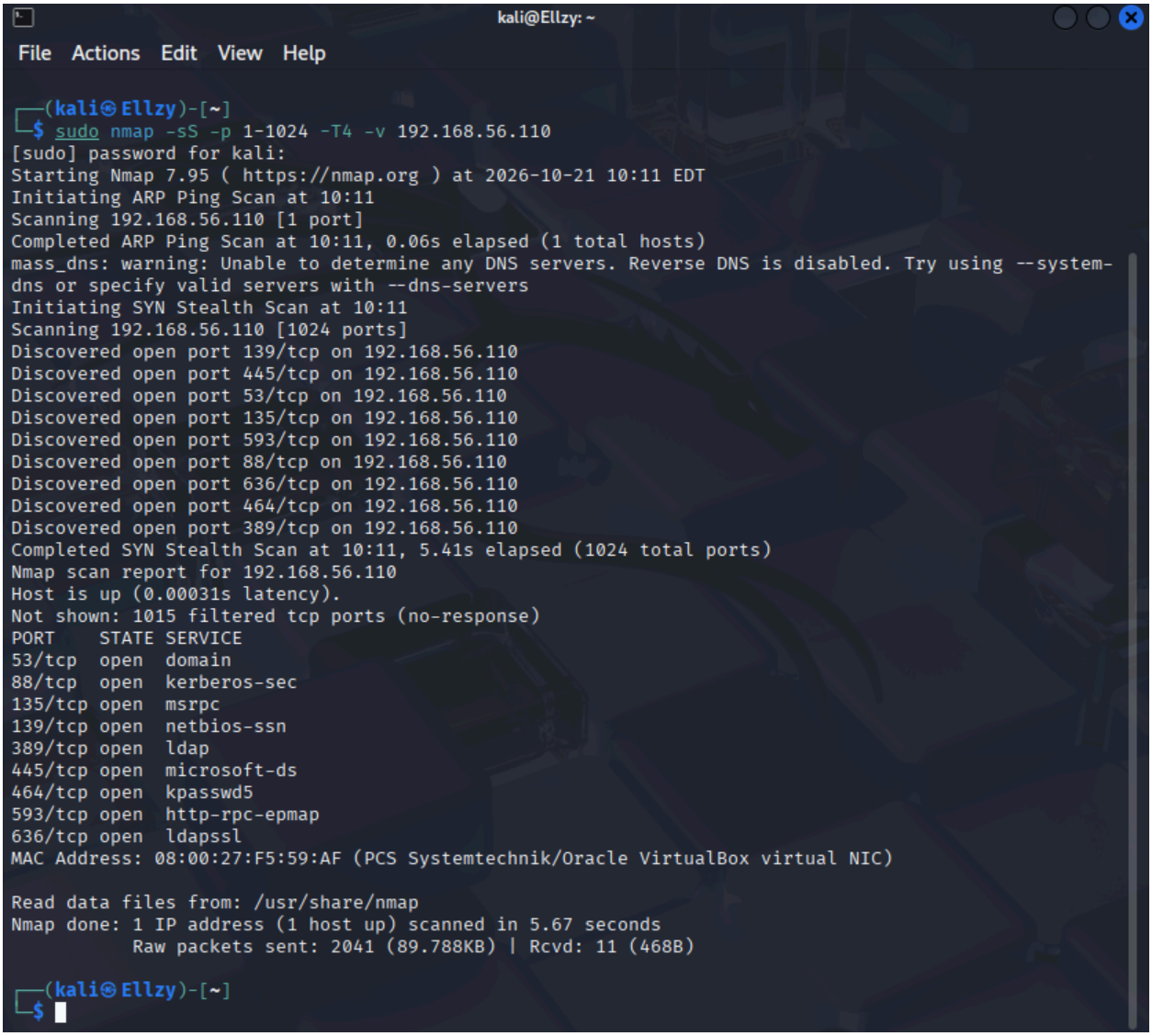
To generate malicious network telemetry, a targeted reconnaissance sweep was executed from Kali Linux against the Domain Controller (DC01). To evade standard application-layer logging (e.g., Windows Event Viewer, IIS logs), a **Half-Open TCP SYN Scan** was utilized.

Execution:

Bash

```
sudo nmap -sS -p 1-1024 -T4 -v 192.168.56.110
```

By manipulating the TCP handshake, the attacker sends only the initial `SYN` packet. When the target replies with `SYN-ACK`, the attacker abruptly sends a `RST` (Reset) packet. Because the session never reaches the `ESTABLISHED` state, the Windows kernel drops the socket connection before passing it to the application layer, evades default logging native OS audit logs.



Phase 2: Protocol Forensics & Packet Dissection

To prove the evasion mechanics at Layer 4, the raw traffic was captured via `tcpdump` on the Ubuntu sensor's promiscuous interface and analyzed in Wireshark.

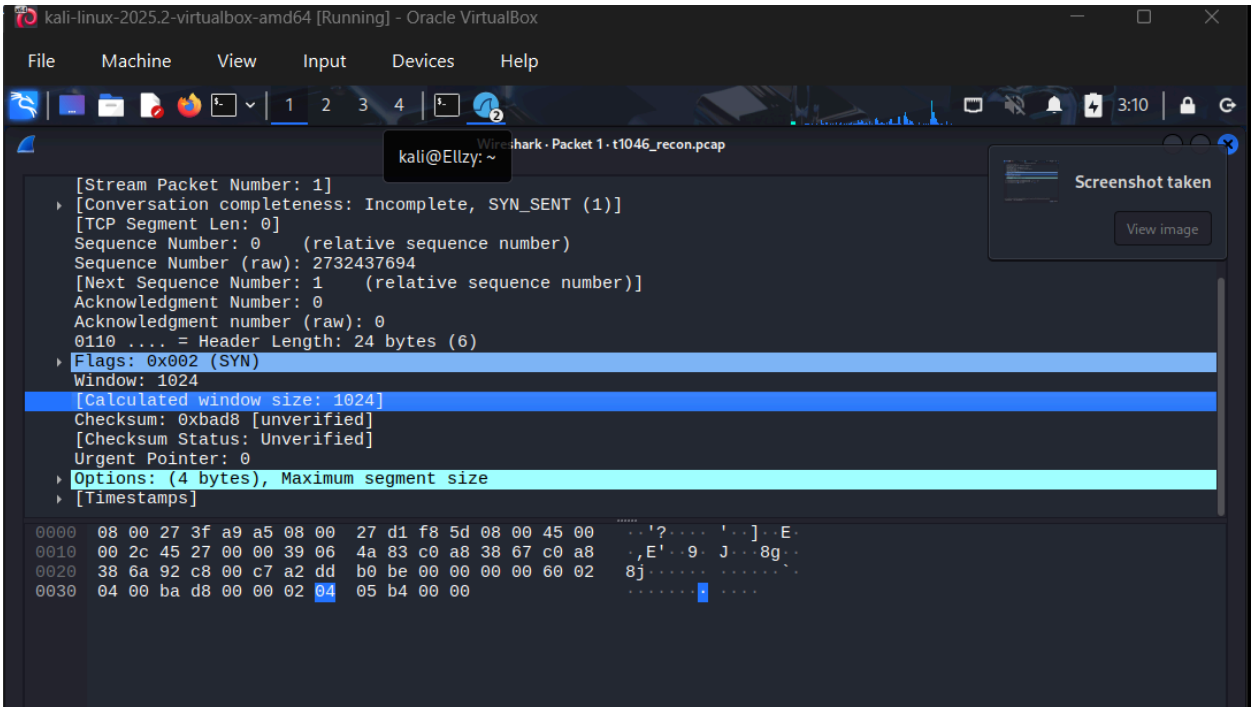
Wireshark Display Filter:

Plaintext

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

Forensic Findings:

- **Flag Manipulation:** The packet headers confirmed `Flags: 0x002 (SYN)` with the `Acknowledgment: Not set`.
- **Connection State:** Wireshark's TCP dissector flagged the streams as `[Conversation completeness: Incomplete, SYN_SENT (1)]`.
- **Behavioral Anomaly:** The capture revealed dozens of isolated `SYN` requests originating from a single IP (`192.168.56.103`) targeting sequential ports on `192.168.56.110` within milliseconds—a clear indicator of automated port enumeration.



Phase 3: Custom Detection Engineering

While Suricata successfully logged the raw flow data, default signature sets (like Emerging Threats) are typically optimized for payload analysis and exploit delivery, meaning raw SYN packets often bypass default alert triggers.

To detect this volumetric anomaly, a custom threshold-based signature was authored in `local.rules` to track connection attempt velocity.

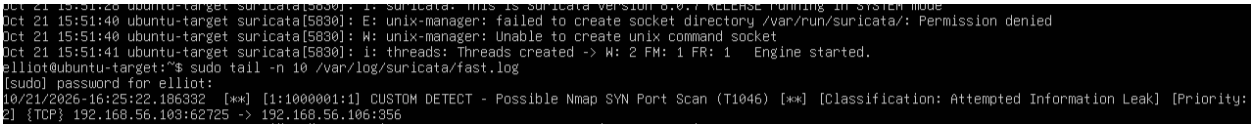
Suricata Rule (`SID: 1000001`):

Plaintext

```
alert tcp any any -> $HOME_NET any (msg:"CUSTOM DETECT - Possible Nmap SYN Port Scan (T10
46)"; flags:S,12; threshold:type both, track by_src, count 20, seconds 5; classtype:attem
pted-recon; sid:1000001; rev:1;)
```

Detection Logic:

- "The rule uses `flags:S,12` to match TCP SYN packets while masking the two reserved bits, targeting raw packets rather than established sessions. The threshold tracks the volume of matching packets—triggering an alert after 20 packets—rather than counting unique destination ports. Note: High-volume legitimate applications or misconfigured network devices generating rapid SYN packets could trigger false positives."



Phase 4: Dual-SIEM Correlation & Threat Hunting

With the detection engine validated, telemetry was piped into two distinct SIEM architectures to demonstrate both signature-driven alerting (Wazuh) and statistical threat hunting (Splunk).

```
("192.168.56.103") (source="/var/log/suricata/eve.json" OR source="Security.xml") | stats
count by source, event_type
```

4.1 Wazuh Pipeline Integration

The Wazuh Agent was configured via `<localfile>` in `ossec.conf` to continuously ingest Suricata's Extensible Event Format (`eve.json`). Wazuh's native decoders automatically parsed the JSON arrays, providing a low-latency, centralized dashboard for immediate triage.



4.2 Splunk Analytics & SPL Threat Hunting

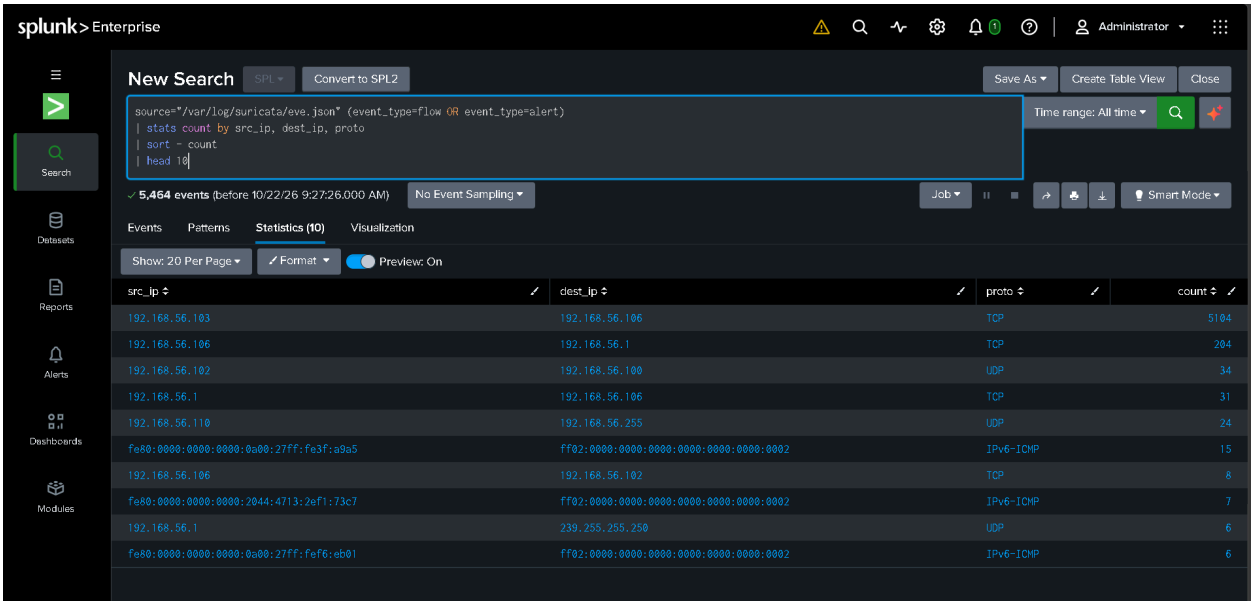
Splunk Enterprise was deployed via Docker directly on the Ubuntu sensor. Telemetry from both the network (Suricata EVE) and the endpoint (Windows Security and Sysmon XML exports) were indexed. Three production-grade hunting queries were executed using Search Processing Language (SPL).

Hunt 1: Network "Top Talkers" (Volumetric Reconnaissance)
To isolate the attacker's IP from background network noise, flow telemetry was aggregated to identify hosts generating disproportionate connection attempts.

Splunk SPL

```
source="/var/log/suricata/eve.json" (event_type=flow OR event_type=alert)
| stats count by src_ip, dest_ip, proto
| sort - count
| head 10
```

- **Finding:** Identified a high volume of inbound TCP SYN packets from a single source IP, confirming automated port scanning activity.

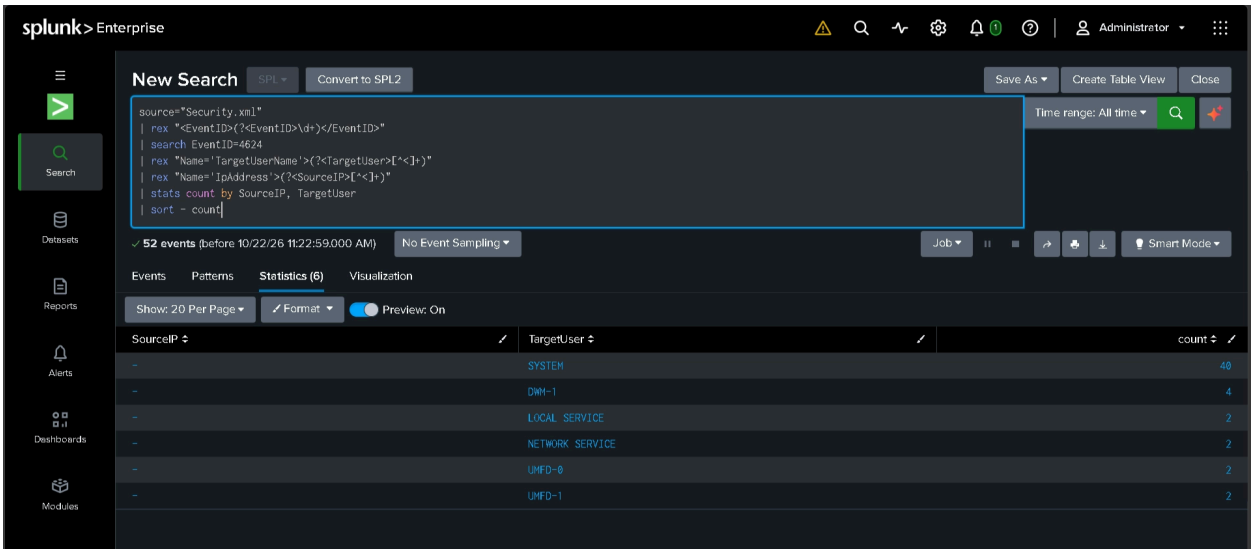


Hunt 2: Logon Activity Tracking (Credential Abuse)
Pivoting from network data to endpoint telemetry, Windows Security logs (Event ID 4624/4625) were analyzed to track authentication behavior across the domain, grouping attempts by source IP and target account to identify potential brute-force or password spraying activity.

Splunk SPL

```
source="Security.xml"
| rex "<EventID>(?<EventID>\d+)</EventID>"
| search EventID=4624 OR EventID=4625
| rex "Name='TargetUserName'>(?<TargetUser>[^<]+)"
| rex "Name='IpAddress'>(?<SourceIP>[^<]+)"
| stats count by SourceIP, TargetUser, EventID
| sort - count
```

- **Finding:** "Correlated Suricata network alerts with endpoint logs to confirm the malicious payload reached the target application."

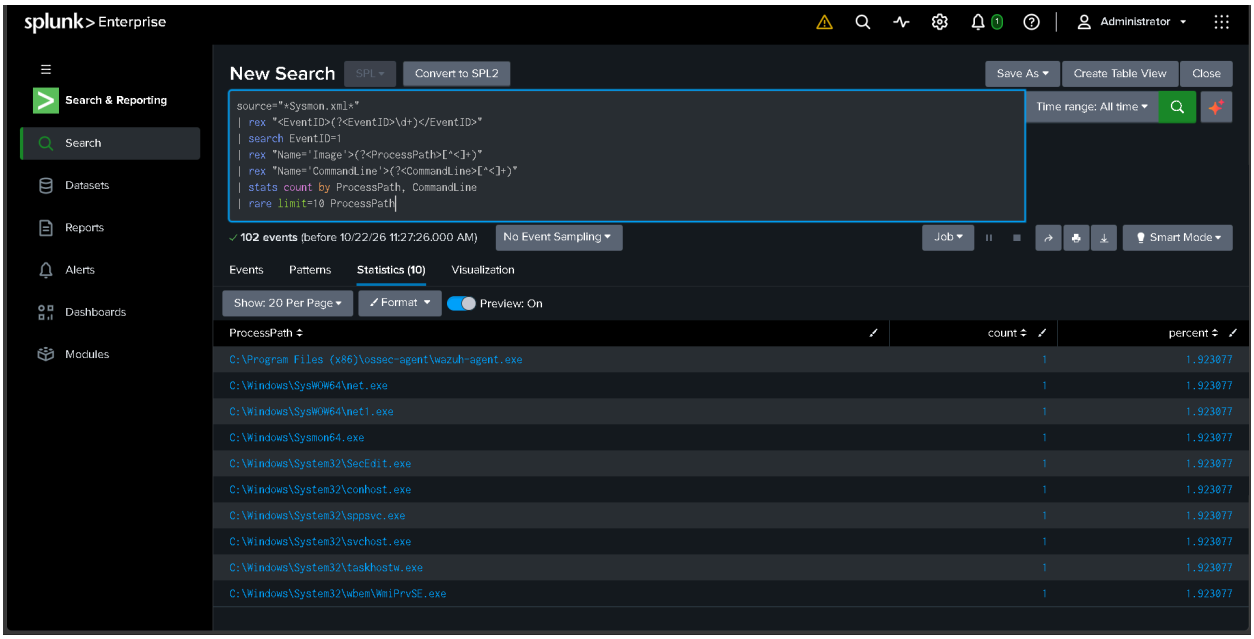


Hunt 3: Rare Processes (Long-Tail Anomaly Detection - T1059)
Because adversaries frequently execute custom payloads or rarely used LOLBins (Living-off-the-Land Binaries), Sysmon Event ID 1 (Process Creation) logs were analyzed using long-tail statistical filtering to isolate outlier executables.

Splunk SPL

```
source="Sysmon.xml"
| rex "<EventID>(?<EventID>\d+)</EventID>"
| search EventID=1
| rex "Name='Image'>(?<ProcessPath>[^<]+)"
| rex "Name='CommandLine'>(?<CommandLine>[^<]+)"
| rex "Name='Image'>(?<ProcessPath>[^<]+)" | rare limit=10 ProcessPath
```

- **Finding:** "Extracted the executed process paths, revealing an anomalous execution sequence spawned by the compromised service."



Engineering Challenges & Root Cause Analysis

Building an enterprise-grade environment on limited virtual hardware required resolving several complex system administration roadblocks:

1. **Linux LVM Disk Exhaustion (Splunk Containerization):**
    - **Symptom:** Docker container extraction failed with `no space left on device`.



    - **Root Cause:** The default Ubuntu LVM installation allocated only 12GB to the root volume (`ubuntu--vg-ubuntu--lv`), leaving the remainder of the virtual disk unallocated.



- **Resolution:** Executed `lvextend -l +100%FREE` and `resize2fs` to dynamically expand the live root filesystem to 23GB without downtime, successfully accommodating Splunk's heavy storage requirements

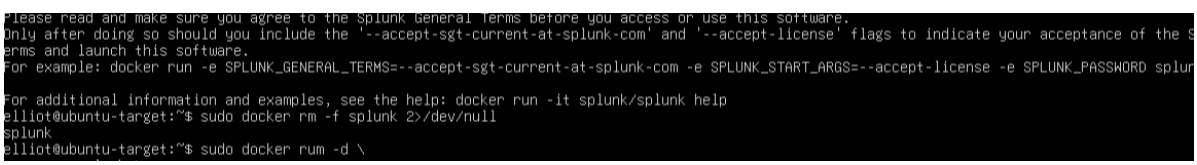




2. **Splunk Headless License Acceptance:**
   - **Symptom:** Splunk Docker container exited immediately with `License not accepted`.
   - **Root Cause:** Newer Splunk Docker image builds implemented a strict requirement for a secondary license flag covering general terms.

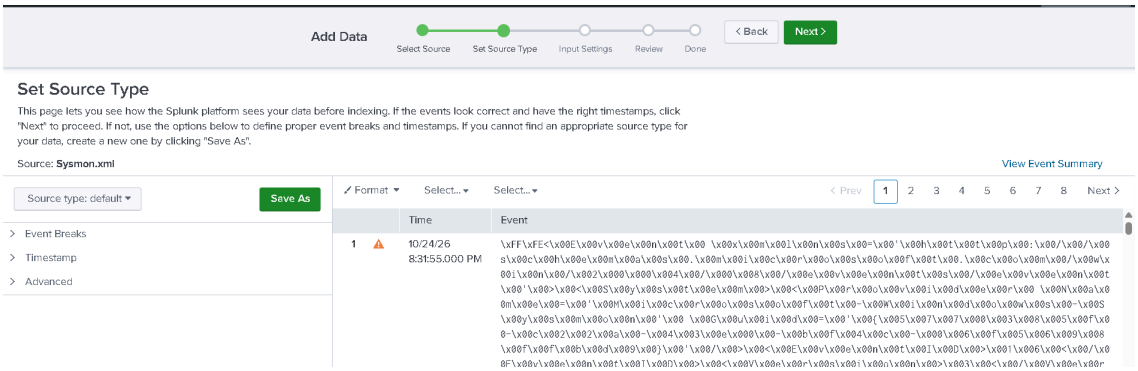


Resolution: I re-deployed the container with the appended environment variable `-e "SPLUNK_GENERAL_TERMS=--accept-sgt-current-at-splunk-com"`.



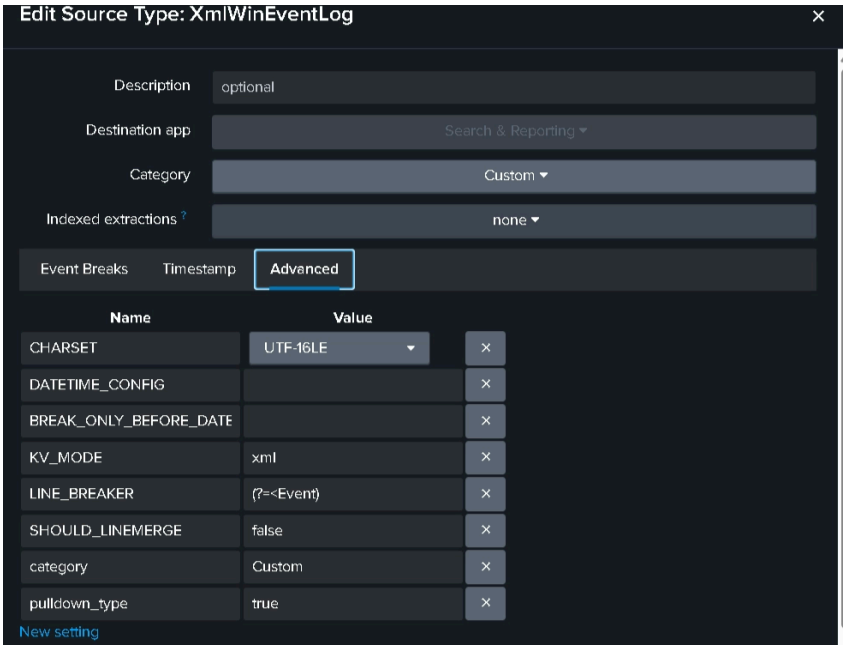
```
elliot@ubuntu-target:~$ sudo docker run -d \
> --name splunk \
> --restart always \
> -p 8000:8000 \
> -e "SPLUNK_GENERAL_TERMS=--accept-sgt-current-at-splunk-com" \
> -e "SPLUNK_START_ARGS=--accept-license" \
> -e "SPLUNK_PASSWORD=EllzySplunk2026!" \
> -v /var/log/suricata:/var/log/suricata:ro \
> splunk/splunk:latest
3840056531b5eb035887153332a1259af31c0b896e3f452aca0a8e0fb0e615a8
elliot@ubuntu-target:~$ _
```

3. **SIEM Binary Decoding Failures (`UTF-16LE`):**
   - **Symptom:** Windows Event Logs (`.evtx`) exported via PowerShell appeared in Splunk as broken binary hex (`\xFF\xFE\x00E...`).



   - **Root Cause:** PowerShell's `Out-File` default encoding utilizes UTF-16LE, which Splunk's Linux backend could not natively parse, treating it as a raw `ElfFile`.
   - **Resolution:** Intercepted the ingestion pipeline at the `props.conf` level via Splunk Web, forcing the `CHARSET` to `UTF-16LE` and configuring `KV_MODE=xml` combined with regex event breaking (`(?=<Event)`) to successfully parse the nested XML schema.

```
[XmlWinEventLog]
CHARSET = UTF-16LE
KV_MODE = xml
LINE_BREAKER = (?=<Event)
```



Incident Response & SOC Triage

1. **Verify the Source:** Inspect the Suricata alert to determine if the scanning IP is external or internal.
2. **Correlate with Endpoint:** Pivot into Splunk using the target's IP to check `Security.xml` or Sysmon logs for successful logons or process executions immediately following the network alert.
3. **Containment:** If endpoint logs show successful exploitation or lateral movement, isolate the compromised host. Block the scanning source IP at the perimeter firewall.

False Positives & Tuning

- **Vulnerability Scanners:** Authorized internal enterprise vulnerability scanners (e.g., Nessus, Qualys) will trigger these network alerts.
- **Tuning Strategy:** Add BPF (Berkeley Packet Filter) exclusions in Suricata or create suppression rules for known, authorized scanner IP addresses.

Strategic Mitigations

- **Network Segmentation:** Place externally facing services in an isolated DMZ. Ensure strict firewall rules deny initiation of traffic from the DMZ into the internal network.
- **Service Hardening:** Disable unnecessary ports and services on internet-facing servers to reduce the available attack surface.

Lessons Learned

- **Log Parsing:** Properly configuring `props.conf` is critical for ensuring JSON network logs (like `eve.json`) are parsed correctly in Splunk, preventing expensive search-time field extractions.
- **Network vs. Endpoint Visibility:** Network alerts provide the "how" (the delivery mechanism), but endpoint logs are required to determine the "what" (whether the attack actually succeeded).

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